

Module 04 Lab - Exploratory Data Analysis

Objective: To learn how to explore a dataset to find patterns, anomalies, and insights before modeling. EDA is like being a detective; you are looking for clues in the data that will help you build a better model.

This lab is fully coded. Your task is to run each cell, read the detailed explanations, understand the purpose of each visualization, and then complete the experimentation section.

Part 1: Setup and Data Loading

What is Exploratory Data Analysis (EDA)?

EDA is the process of using summary statistics and visualizations to understand a dataset's main characteristics. Before you can build a model, you need to understand your data. What stories does it tell? Are there errors or missing values? Are there strong relationships between variables? EDA helps answer these questions.

We will use the famous Titanic dataset for this lab. It contains information about passengers and, crucially, whether they survived the disaster.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns # Seaborn is a library built on top of Matplotlib that makes creating beautiful plots easier.

# Load the dataset directly from a URL
df = pd.read_csv('https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv')

print("--- First 5 Rows ---")
print(df.head())

print("\n--- Basic Info ---")
# .info() is a great first command. It tells us the column names, how many non-null values are in each column, and their data types.
# Notice that 'Age' and 'Cabin' have missing values!
df.info()
```

```
--- First 5 Rows ---
  PassengerId  Survived  Pclass \
0             1         0       3
1             2         1       1
2             3         1       3
3             4         1       1
4             5         0       3

      Name               Sex  Age  SibSp \
0      Braund, Mr. Owen Harris    male  22.0      1
1  Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0      1
2      Heikkinen, Miss. Laina    female  26.0      0
3  Futrelle, Mrs. Jacques Heath (Lily May Peel)    female  35.0      1
4      Allen, Mr. William Henry    male  35.0      0
```

```
   Parch  Ticket   Fare Cabin Embarked
0      0   A/5 21171   7.2500   NaN      S
1      0    PC 17599  71.2833   C85      C
2      0  STON/O2. 3101282   7.9250   NaN      S
3      0   113803  53.1000  C123      S
4      0   373450   8.0500   NaN      S
```

```
--- Basic Info ---
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column      Non-Null Count  Dtype
---  -
0   PassengerId  891 non-null    int64
1   Survived     891 non-null    int64
2   Pclass       891 non-null    int64
3   Name         891 non-null    object
4   Sex          891 non-null    object
5   Age         714 non-null    float64
6   SibSp        891 non-null    int64
7   Parch        891 non-null    int64
8   Ticket       891 non-null    object
9   Fare         891 non-null    float64
10  Cabin        204 non-null    object
11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

Part 2: Descriptive Statistics

Let's start by getting a high-level numerical summary of the data. The `.describe()` method is perfect for this. It calculates statistics like mean, standard deviation, min, and max for the numerical columns.

```
# Get summary statistics for numerical columns
print("--- Descriptive Statistics ---")
print(df.describe())
```

```
print("\n--- Key Insights from Statistics ---")
print(f"The average age of a passenger was {df['Age'].mean():.1f} years.")
print(f"The overall survival rate was {df['Survived'].mean():.1%}.")
print(f"Fares ranged from ${df['Fare'].min()} to a whopping ${df['Fare'].max()}."
```

```
--- Descriptive Statistics ---
PassengerId  Survived  Pclass    Age    SibSp  \
count  891.000000  891.000000  891.000000  714.000000  891.000000
mean    446.000000    0.383838    2.308642   29.699118    0.523008
std     257.353842    0.486592    0.836071   14.526497    1.102743
min       1.000000    0.000000    1.000000    0.420000    0.000000
25%     223.500000    0.000000    2.000000   20.125000    0.000000
50%     446.000000    0.000000    3.000000   28.000000    0.000000
75%     668.500000    1.000000    3.000000   38.000000    1.000000
max     891.000000    1.000000    3.000000   80.000000    8.000000
```

```

Parch  Fare
count  891.000000  891.000000
mean    0.381594   32.204208
std     0.806057   49.693429
min     0.000000    0.000000
25%     0.000000    7.910400
50%     0.000000   14.454200
75%     0.000000   31.000000
max     6.000000  512.329200
```

```
--- Key Insights from Statistics ---
The average age of a passenger was 29.7 years.
The overall survival rate was 38.4%.
Fares ranged from $0.0 to a whopping $512.3292.
```

Part 3: Visual EDA - Telling Stories with Plots

Numbers are great, but plots make patterns and relationships immediately obvious. The goal of visual EDA is to turn data into insights.

A Note on Plotting Libraries:

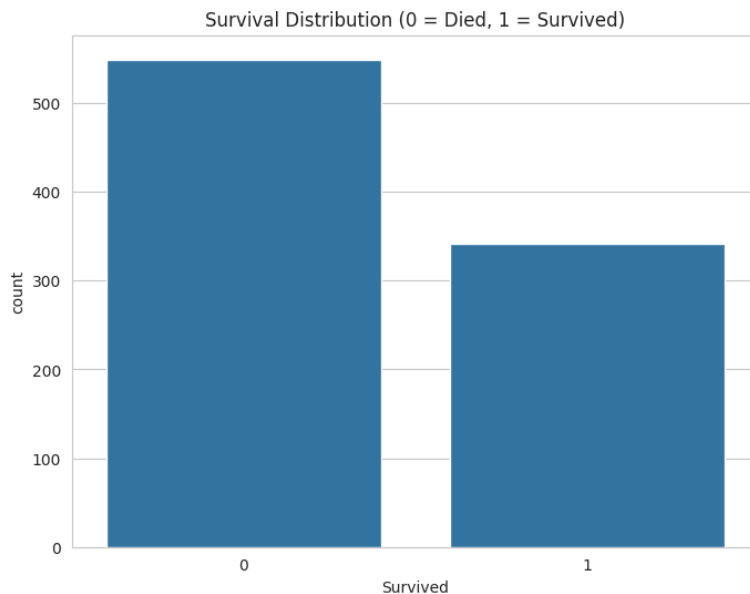
- **Matplotlib:** The foundational library, gives you full control over everything.
- **Seaborn:** Built on Matplotlib, it provides a simpler, high-level interface for creating common statistical plots. We will use Seaborn for its ease of use and attractive defaults.

Visualization 1: How many survived?

A simple `countplot` is the best way to see the distribution of a categorical variable, like our target `Survived`.

```
sns.set_style('whitegrid') # Sets a nice visual style for our plots
plt.figure(figsize=(8, 6))
sns.countplot(x='Survived', data=df)
plt.title('Survival Distribution (0 = Died, 1 = Survived)')
plt.show()
```

```
print("Insight: Far more people died than survived. This is an example of an imbalanced dataset, which can sometimes be a challenge for machine learning models.
```



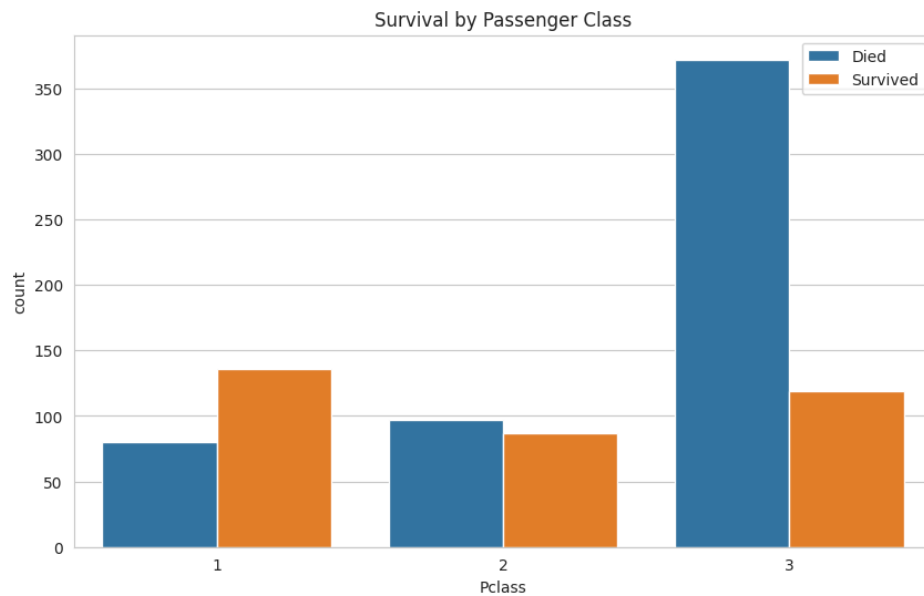
```
Insight: Far more people died than survived. This is an example of an imbalanced dataset, which can sometimes be a challenge for machine learning models.
```

Visualization 2: Does passenger class matter for survival?

Now we want to see if there's a relationship between two variables. We can use the `hue` parameter in `countplot` to split the bars by another category.

```
plt.figure(figsize=(10, 6))
sns.countplot(x='Pclass', hue='Survived', data=df)
plt.title('Survival by Passenger Class')
plt.legend(['Died', 'Survived'])
plt.show()
```

print("Insight: This is a very strong pattern. 1st class passengers had a much higher chance of survival compared to 3rd class passengers. Money seems to have made a difference.")



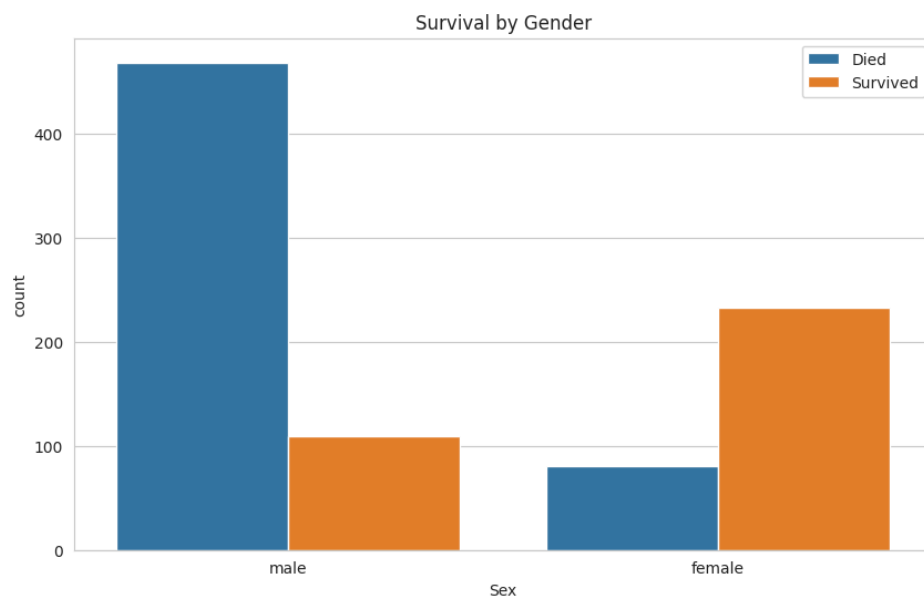
Insight: This is a very strong pattern. 1st class passengers had a much higher chance of survival compared to 3rd class passengers. Money seems to have made a difference.

Visualization 3: What about gender?

The 'women and children first' mantra is famous. Let's see if the data supports it.

```
plt.figure(figsize=(10, 6))
sns.countplot(x='Sex', hue='Survived', data=df)
plt.title('Survival by Gender')
plt.legend(['Died', 'Survived'])
plt.show()
```

print("Insight: The pattern is undeniable. A much higher proportion of females survived compared to males. This is another very strong predictor.")



Insight: The pattern is undeniable. A much higher proportion of females survived compared to males. This is another very strong predictor.

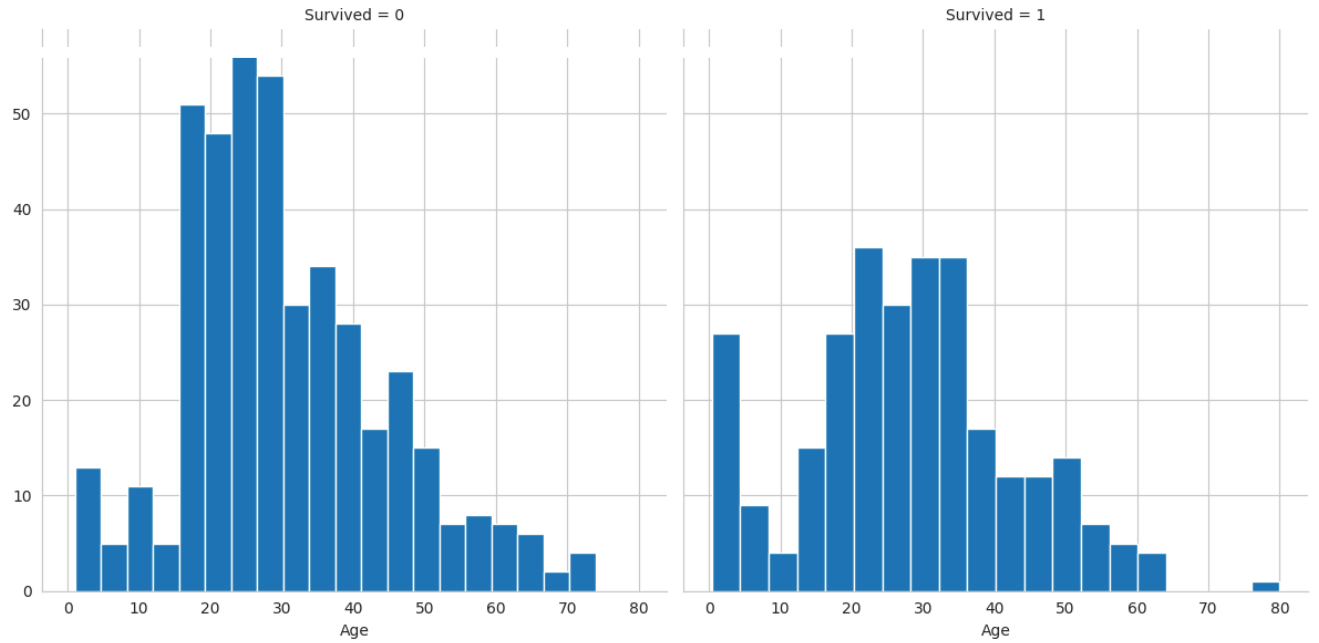
Visualization 4: How does age play a role?

For a continuous variable like `Age`, a histogram is a great way to see its distribution.

```
# A FacetGrid allows us to create multiple plots side-by-side to compare distributions.
# Here, we create one histogram for passengers who died (col='Survived'=0) and one for those who survived (col='Survived'=1).
g = sns.FacetGrid(df, col='Survived', height=6)
```

```
g.map(plt.hist, 'Age', bins=20)
plt.show()
```

```
print("Insight: The age distribution for those who did not survive is centered around the 20-40 age range. For those who survived, there is a noticeable spike for y
```



```
Insight: The age distribution for those who did not survive is centered around the 20-40 age range. For those who survived, there is a noticeable spike for y
```

Part 4: Student Experimentation

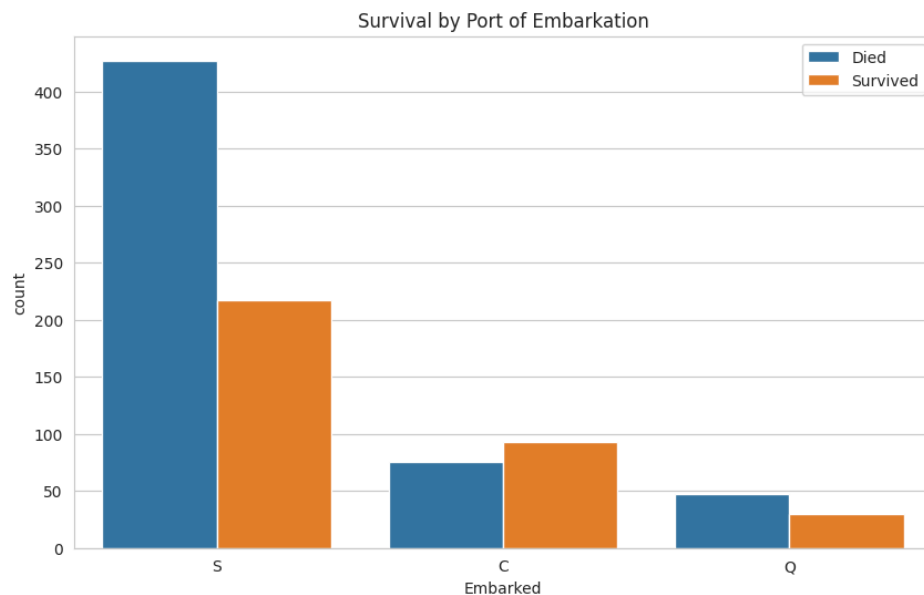
Instructions: Create your own visualizations to explore other relationships in the data.

Experiment 1: Port of Embarkation

1. The `Embarked` column tells you where a passenger boarded the ship (C = Cherbourg, Q = Queenstown, S = Southampton).
2. Create a `countplot` to see how survival rates differed by the port of embarkation. Does where they boarded seem to be related to their survival?

```
plt.figure(figsize=(10, 6))
sns.countplot(x='Embarked', hue='Survived', data=df)
plt.title('Survival by Port of Embarkation')
plt.legend(['Died', 'Survived'])
plt.show()
```

```
print("Insight: The port of embarkation is related to survival.\nCherbourg is the only port where more passengers survived than died, while Southampton and Q
```



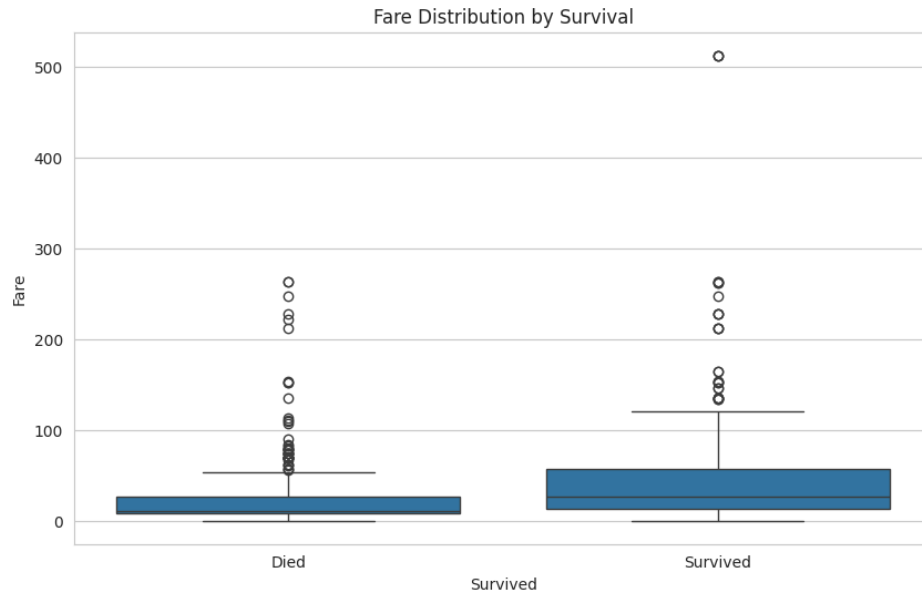
```
Insight: The port of embarkation is related to survival.
Cherbourg is the only port where more passengers survived than died, while Southampton and Queenstown both had more deaths.
```

Experiment 2: Fare vs. Survival

1. `Fare` is a continuous numerical variable.
2. A `boxplot` or `violinplot` is a great way to see the distribution of fares for those who survived vs. those who didn't.
3. Create one of these plots to compare the `Fare` distribution by `Survived`. What does it tell you?

```
plt.figure(figsize=(10, 6))
sns.boxplot(x='Survived', y='Fare', data=df)
plt.title('Fare Distribution by Survival')
plt.xticks([0, 1], ['Died', 'Survived'])
plt.show()
```

```
print("Insight: Survivors generally paid higher fares, which matches the passenger class pattern seen earlier.")
```



Insight: Survivors generally paid higher fares, which matches the passenger class pattern seen earlier.

Knowledge Check

Instructions: Answer the following questions in this markdown cell.

1. **What is the primary goal of Exploratory Data Analysis (EDA)?**
2. **Based on the plots in this lab, what kind of person had the best chance of surviving the Titanic?** (Describe them in terms of class, gender, and age).
3. **Why is it important to visualize data instead of just looking at summary statistics?** What can a plot show you that a number like 'mean' or 'count' can't?

Q1 Answer: The primary goal of EDA is to understand the main characteristics of a dataset beforehand, to ensure a model is properly built. Missing values or errors, relationships between variables, and other anomalies must be thoroughly examined.

Q2 Answer: Based on the plots, first-class female passengers had the best chance of surviving the Titanic. Children under 10 also showed a noticeably higher survival rate in the age distribution histogram.

Q3 Answer: Visualizing data better demonstrates the differences and relationships within a dataset. For example, the mean fare doesn't reveal the skew, the outliers, or the fact that most passengers clustered at the low end, but the boxplot shows all three at once.